



Breast Cancer Detection in Mammograms Through Vision Transformer with Enhanced Preprocessing and Attention-Driven Visualization

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DECLARATION

I hereby declare that the work herein, now submitted as a thesis report for the degree of Master's In Data Science at Charles Darwin University, is the result of my own investigations, and all references to ideas and work of other researchers have been specifically acknowledged. I hereby certify that the work embodied in this thesis report has not already been accepted in substance for any degree, and is not being currently submitted in candidature for any other degree.

Signature: Raj Kumar Sah

Date: 01 June 2025

DEDICATION

This project is dedicated to all individuals and families affected by breast cancer. Their strength and resilience continue to inspire advancements in medical research and technology. We also extend our most profound appreciation to the radiologists, clinicians, and data scientists whose dedication to early detection and diagnosis has laid the foundation for this work. Lastly, we dedicate this research to our mentors and families, whose unwavering support made this endeavor possible.

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Additionally, I acknowledge the authors and contributors of the VinDr-Mammo dataset, whose publicly available annotations and imaging data were critical to this study. The use of such datasets, along with open-source research tools and libraries, greatly facilitated the development of this work.

Finally, I am grateful to my family and friends for their unwavering support, patience, and motivation throughout this journey.

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LIST OF SYMBOLS

(required if you use symbols)¹

¹ List here all variables and their units used in the paper and other sections of this document, remove this footnote

Breast Cancer Detection in Mammograms Through Vision Transformer with Enhanced Preprocessing and Attention-Driven Visualization

Abstract—Breast cancer is a primary global health concern, with mammography playing a crucial role in early detection. However, accurately interpreting mammograms continues to be difficult due to subtle lesion characteristics and issues with image quality. This research introduces an advanced and interpretable deep-learning framework that utilizes Vision Transformer (ViT) combined with ResNet50 for feature extraction and sophisticated preprocessing and augmentation techniques. Utilizing the VinDr-Mammo dataset of 5,000 mammograms, we applied wavelet denoising, CLAHE, unsharp masking, and muscle removal to enhance the images. We addressed class imbalance by generating synthetic images using CycleGAN, MixUp, and CutMix augmentations. Our ViT model achieved an AUC of 0.91 with an optimal diagnostic threshold of 0.49. Attention rollout visualizations emphasized clinically relevant areas in the images, significantly enhancing interpretability. This methodology provides promising support for clinical mammography.

Keywords—Breast Cancer, Mammogram, Vision Transformer, AUC.

I. INTRODUCTION

Breast cancer continues to be the leading cause of cancer-related deaths among women worldwide, with millions of diagnoses each year [1]. Early and precise diagnosis through mammography greatly enhances survival rates by facilitating timely treatment options [2]. Nonetheless, interpreting mammograms manually is inherently difficult because of subtle tumor features, dense breast tissues, and considerable variations in image quality [3]–[5].

Deep learning (DL) methods, particularly Convolutional Neural Networks (CNNs), have revolutionized medical imaging diagnostics by automating and enhancing the accuracy of mammogram interpretation [6]–[9]. Despite these advances, CNN-based methods often encounter limitations in interpretability, hindering clinical trust and adoption [10], [11]. Recently, Vision Transformer (ViT) architectures have emerged, offering promising capabilities in medical image analysis due to their intrinsic attention mechanisms that provide transparent insights into model decisions [12], [13].

This research presents a new framework that combines advanced preprocessing techniques, ViT architectures,

and attention-based visualization methods to enhance diagnostic accuracy and interpretability in breast cancer detection from mammograms. We specifically address image quality and class imbalance issues, improving diagnostic transparency through attention visualization methods, including attention rollout and Grad-CAM [14]–[16]. This study contributes significantly to the literature by offering interpretable deep learning solutions with clinical potential for integration.

Background

Breast cancer ranks as one of the leading causes of cancer-related deaths in women, highlighting the necessity of early and precise diagnosis [1]. Full-field digital mammography (FFDM) is commonly utilized for screening; however, its interpretation is often complicated by dense tissue overlaps, subtle lesions, and variability among observers [2], [3]. Traditional rule-based computer-aided detection (CAD) systems have shown only modest improvements in diagnostic accuracy, leading to an increased focus on deep learning methods that can identify distinguishing features directly from raw image data [4].

Convolutional Neural Networks (CNNs) have demonstrated strong performance in medical imaging tasks like lesion classification and segmentation. Still, they often struggle to capture global contextual relationships within large images such as mammograms [5], [6]. Vision Transformers (ViTs) have recently emerged as an alternative, using self-attention mechanisms to model long-range dependencies across image patches [7]. This makes them particularly suitable for analyzing complex visual patterns in high-resolution mammograms. When paired with attention-guided learning, these models enhance accuracy and provide interpretable predictions—an essential factor for clinical trust [8].

Despite progress, class imbalance and limited data availability hinder model generalization. Generative Adversarial Networks (GANs), particularly CycleGANs, have been employed to synthetically generate mammograms and address this issue by enhancing the representation of under-sampled malignant classes [9], [10]. Combining such augmentation techniques with ViT-based attention modeling presents a promising direction for constructing scalable, robust, and interpretable AI-assisted diagnostic pipelines.

Aim of Research

Accurate interpretation of mammograms remains a critical but difficult task due to the complex anatomy of breast tissue, subtle differences between benign and malignant findings, and class imbalance in real-world data. Additionally, a lack of interpretability in AI models limits clinical trust and adoption. This research addresses these challenges by integrating deep learning and attention-based techniques to enhance diagnostic accuracy, support radiologists, and enable scalable breast cancer screening.

The aim of this research is to:

Improve Diagnostic Accuracy

Utilize Vision Transformers (ViTs) to model global context in mammogram images for improved classification between benign and malignant cases.

Handle Class Imbalance

Augment the dataset using classical methods and CycleGAN to enhance minority class representation, improving model robustness.

Enable Interpretability

Apply Grad-CAM and attention rollout to visualize prediction logic, fostering transparency and trust in clinical settings.

Structure of Paper

II. APPROACH

Advanced Image Preprocessing

Wavelet Denoising:- Wavelet-based denoising reduces noise while preserving edges and delicate structures. The mathematical formulation involves thresholding wavelet coefficients:

$$I_{denoised} = W^{-1}(T(W(I)))$$

Where:

W : wavelet transform

T : thresholding function (e.g., soft or hard thresholding)

W^{-1} : inverse wavelet transform

I : original image

```
def wavelet_denoise(image, wavelet='db1', level=2):
    coeffs = pywt.wavedec2(image, wavelet, level=level)
    coeffs_thresh = [pywt.threshold(c, 20, 'soft') for c in coeffs[1:]]
    return pywt.waverec2([coeffs[0]] + coeffs_thresh, wavelet)
```

CLAHE (Contrast Limited Adaptive Histogram Equalization):- CLAHE improves local contrast by limiting histogram amplification to avoid noise over-enhancement:

$$s = (L - 1) \cdot \sum_{j=0}^r \frac{\rho_r(r_j)}{MN}$$

Where:

L : number of gray levels

ρ_r : histogram probability of intensity r_j

$M \times N$: image dimensions

```
def apply_clahe(image):
    clahe = cv2.createCLAHE(clipLimit=2.0,
    tileGridSize=(8,8))
    return clahe.apply(image)
```

Unsharp Masking:- Enhances image sharpness by adding a scaled difference between the original and a blurred image:

$$I_{sharp} = I + \alpha(I - G(I))$$

Where: I : input image

$G(I)$: Gaussian-blurred image

α : sharpening factor

```
def unsharp_mask(image, alpha=1.5):
    blurred = cv2.GaussianBlur(image, (5, 5), 0)
    return cv2.addWeighted(image, 1 + alpha, blurred,
    -alpha, 0)
```

Muscle and Background Removal:- Removes non-breast areas using morphological and intensity-based masking. Steps:

- Binary thresholding
- Morphological operations (e.g., dilation/erosion)
- Largest contour filtering\

```
def remove_background(image):
    _, thresh = cv2.threshold(image, 20, 255,
    cv2.THRESH_BINARY)
    contours, _ = cv2.findContours(thresh,
    cv2.RETR_EXTERNAL, cv2.CHAIN_APPROX_SIMPLE)
    largest = max(contours, key=cv2.contourArea)
    mask = np.zeros_like(image)
    cv2.drawContours(mask, [largest], -1, 255, -1)
    return cv2.bitwise_and(image, image, mask=mask)
```

Synthetic Image Generation

To mitigate the class imbalance problem inherent in mammogram datasets, this study incorporates three advanced data augmentation strategies—CycleGAN, MixUp, and CutMix—each leveraging unique mechanisms to generate diverse training data and enhance generalization.

- **CycleGAN:** CycleGAN enables unpaired image-to-image translation and is particularly effective in generating synthetic mammogram images that simulate underrepresented abnormal patterns.

Loss Function:

$$L_{total} = L_{GAN}(G, D_Y, X, Y) + L_{gan}(F, D_X, Y, X) + \lambda L_{cycle}(G, F)$$

Where

- G, F : Generator networks
- D_X, D_Y : Discriminator networks

- L_{cycle} : Cycle-consistency loss ensuring $F(G(x)) \approx x$
- λ : Hyperparameter balancing losses

This technique generates realistic synthetic images that help rebalance the dataset.

- **MixUp**:- MixUp is a regularization technique that blends two images and their labels to create a new virtual training sample. This forces the model to learn smoother decision boundaries rather than memorize sharp class distinctions.

$$\bar{x} = \lambda x_i + (1 - \lambda)x_j$$

$$\bar{y} = \lambda y_i + (1 - \lambda)y_j$$

Where: $\lambda \sim \text{Beta}(\alpha, \alpha)$, usually with $\alpha = 0.2$

- **CutMix**: CutMix improves on MixUp by cutting and pasting patches between images rather than linearly blending the entire image. It helps the model learn localized features and reduces overdependence on a single region.

$$\bar{x} = x_i \cdot 1_M + x_j \cdot (1 - 1_M)$$

$$\bar{y} = \lambda y_i + (1 - \lambda)y_j$$

Where: 1_M : Binary mask for the patch

λ : Ratio based on the area of the mask

Feature Extraction Using ResNet50

Feature extraction is crucial in medical image processing, converting pixel-level data into structured, high-dimensional feature embeddings. In this study, we utilize ResNet50, a 50-layer deep convolutional neural network, to extract features from mammogram images before classification with the Vision Transformer.

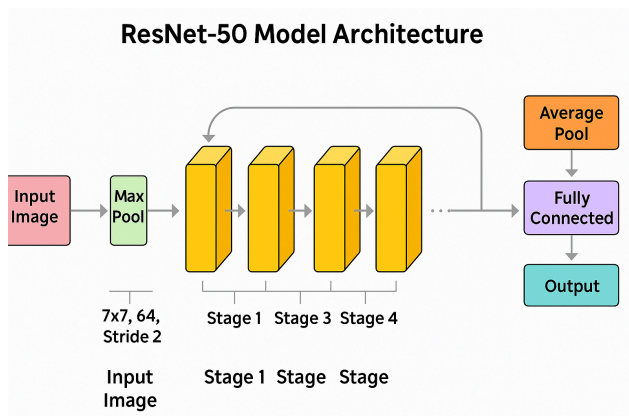


Figure 1: ResNet50 Model Architecture

ResNet50 uses residual connections to ease the training of deep networks by directly passing information across layers. The primary formulation of a residual block is:

$$y = F(x, W_i) + x$$

Where

- x : input feature map
- $F(x, W_i)$: residual function with learnable parameters W_i ,
- y : output of the block

The residual mapping $F(x)$ can be explicitly expanded as:

$$F(x) = W_2 \cdot \text{RELU}(W_1 \cdot x + b_1) + b_2$$

This architecture enhances gradient flow in deep networks, enabling effective learning of both low-level and high-level feature representations. This capability in mammography is crucial for identifying overall breast tissue patterns and localized lesion textures.

In this implementation, we utilized a ResNet50 model that was pretrained on the ImageNet dataset. We removed the classification head and retained the global average pooling layer output as a feature vector with 2048 dimensions. This vector encapsulates the spatial and semantic features of the mammogram image.

The feature vector is subsequently fed into the Vision Transformer module, merging ResNet50's local pattern recognition strengths with the global attention mechanisms of ViT. This combined approach enhances diagnostic accuracy in detecting intricate pathological features.

Classification Using Vision Transformer (ViT)

The Vision Transformer (ViT) is a deep learning architecture based on transformers that was initially designed for natural language processing and subsequently modified for image recognition tasks [28]. Unlike CNNs that depend on localized receptive fields, ViT considers an image as a sequence of fixed-size patches, utilizing self-attention to understand global relationships throughout the entire image. This approach is especially beneficial in mammography, where identifying long-range dependencies, like asymmetry or scattered calcifications, is vital for diagnosis.

Why ViT for Mammography?

Convolutional Neural Networks (CNNs) face inherent limitations in capturing non-local features because of their restricted receptive fields. In contrast, the Vision Transformer (ViT) employs a global self-attention mechanism that allows it to assess interactions across the whole image, making it particularly effective in identifying widely distributed lesions and global structural abnormalities typical in breast cancer cases [29]. Additionally, ViT provides inherent interpretability through attention maps, enabling radiologists to see which regions affected the model's decisions.

Vision Transformer Pipeline and Equations

- **Patch Embedding:** Given an image $x \in \mathbb{R}^{H \times W \times C}$, it is divided into $N = \frac{HW}{P \times P}$ non-overlapping patches of a size $P \times P$. Each patch is flattened into a vector:

$$x_p^i \in \mathbb{R}^{P^2 \cdot C}, \text{ for } i = 1, 2, 3, \dots, N$$

- **Linear Projection and Positional Encoding:** Each flattened patch is projected to a D-dimensional embedding, and a learnable [CLS] token is prepended. Positional encodings are added:

$$z_0 = [x_{class}; x_p^1 E; x_p^2 E; \dots; x_p^N E] + E_{pos}$$

Where

$$E \in \mathbb{R}^{(P^2 \cdot C) \times D}: \text{patch embedding projection}$$

$$E_{pos}: \text{positional encodings}$$

- **Multi-Head Self-Attention(MHSA):** For each transformer encode block, attention is computed as:

$$Attention(Q, K, V) = softmax(\frac{QK^T}{\sqrt{d_k}})V$$

With

$$Q = XW^Q, \quad K = XW^K, \quad V = XW^V$$

Where: W^Q, W^K, W^V are learned projection matrices, and d_k is the dimension of the key vectors.

- **Feed Forward Network(FFN):** Each encoder block includes an FFN applied to each patch embedding:

$$FFN(x) = GELU(xW_1 + b_1)W_2 + b_2$$

- **Classification Head:** The final embedding of the [CLS] token is passed to a linear layer for classification:

$$y = softmax(W_{head} \cdot z_{CLS})$$

ViT Architecture

This architecture enables the model to dynamically focus on the most relevant areas in the mammogram, presented visually through attention rollout and Grad-CAM to enhance clinical interpretability.

The Vision Transformer (ViT) is an innovative deep learning architecture that utilizes self-attention mechanisms on sequences of image patches. This approach enables it to effectively capture global relationships across various parts of an input image. In contrast to CNNs, which depend on local receptive fields, ViTs analyze pairwise interactions among all patches, facilitating the modeling of long-range dependencies essential for recognizing intricate patterns in medical images such as mammograms.

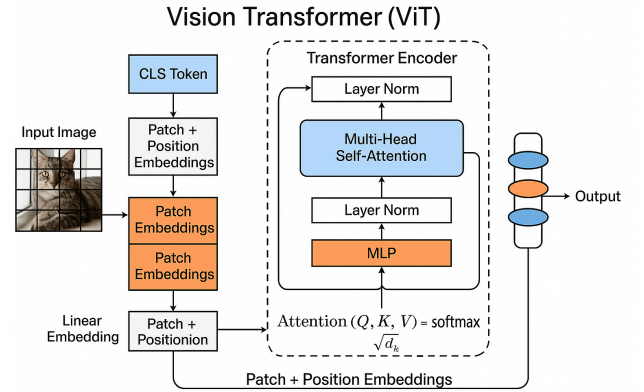


Figure 2: Vision Transformer Architecture

Each input image is segmented into fixed-size patches that are flattened and embedded linearly. These embeddings incorporate positional encodings to preserve spatial information and are then processed through several transformer encoder layers. Each encoder layer includes a Multi-Head Self-Attention (MHSA) block followed by a Feed Forward Network (FFN), with layer normalization and residual connections implemented at each step.

We chose ViT for this study due to its exceptional ability to learn non-local interactions across mammographic regions, which is essential for capturing both overall symmetry and localized irregularities. Additionally, ViT offers a straightforward means of interpretability through attention maps, enhancing the explainability of its predictions for clinical applications.

Interpretability Through Attention Rollout

A key factor in implementing AI-based diagnostic tools in clinical settings is interpretability. In mammography, it's not enough to produce precise predictions; clinicians need to recognize which areas of the image affected the model's outcome. To promote transparency and build clinical trust, we utilize two commonly employed interpretability techniques: Attention Rollout.

Attention Rollout

Each self-attention layer calculates attention scores between every token (image patch) in Vision Transformers. To visualize how input patches impact the final decision, we utilize attention rollout—a method that compiles attention weights across layers.

Let's A_l denote the attention matrix at the transformer layer l . The cumulative attention map $\hat{A}$ across L layers is:

$$\hat{A} = A_1 \cdot A_2 \cdot \dots \cdot A_L$$

This product tracks the accumulation of attention by the [CLS] token throughout the network, emphasizing image areas that significantly impact the classification result. An attention heatmap is then overlaid on the mammogram for enhanced visualization.

Related Work

This section examines foundational and recent research relevant to each key element of our framework, highlighting how our work fits within the broader landscape of medical image analysis and deep learning. It compares current methodologies with the unique contributions of our integrated approach.

Mammogram Image Preprocessing: Enhancing contrast and reducing irrelevant structures by preprocessing mammograms is crucial for improving CAD performance. Wavelet denoising is commonly used to maintain important edges while removing noise [17]. CLAHE effectively improves subtle details within dense breast tissue [18]. While unsharp masking is typically used in photography, it has also been successful in making lesion boundaries more visible in clinical images [19]. Eliminating the pectoral muscle in MLO views helps reduce false positives and improves region-focused analysis [20]. Our study combines all these techniques into a unified, automated preprocessing pipeline aimed at optimizing the quality of visual features.

Addressing Class Imbalance with Data Augmentation: Class imbalance in mammographic datasets poses a significant challenge to model generalization. Prior research has employed CycleGAN to generate synthetic

pathological images without requiring paired datasets [21], [22]. MixUp and CutMix have been applied in both natural and medical imaging tasks, offering label-preserving transformations that improve model robustness [23], [24]. Our advancement includes the amalgamation of all three. Methods—CycleGAN for image-level diversity, MixUp for label smoothing, and CutMix for region-centered blending—specifically tailored to address rare lesion types in mammography.

Deep Feature Extraction with ResNet50: CNN architectures like ResNet50 are still preferred for their hierarchical learning features and residual connections that mitigate vanishing gradients. Previous research has demonstrated its effectiveness in identifying breast masses, classifying calcifications, and distinguishing benign from malignant cases [25], [26]. The trend of using a pretrained ResNet50 as the base for transformer-based classifiers is increasing, as ImageNet pretraining significantly boosts performance in data-scarce areas such as medical imaging [27].

Vision Transformers in Medical Image Analysis: Vision Transformers (ViTs) have demonstrated superior performance compared to Convolutional Neural Networks (CNNs) in several medical imaging benchmarks, particularly in cases where understanding global context is essential. Dosovitskiy et al. [28] were the pioneers of ViT in vision-related tasks, and later research efforts, such as TransUNet [29] and MedViT [30], have tailored transformers for medical use. The capacity of ViTs to focus on spatially separated patches is particularly beneficial in mammography, as malignant patterns can significantly span both sides or show architectural distortions.

Interpretability with Attention-Based Visualization: Attention Rollout, designed for transformer-based architectures, facilitates the visual tracing of decision-making processes across layers [32]. Its effectiveness has been demonstrated in cancer histopathology [33], lung CT diagnosis [34], and, more recently, breast imaging [35]. We utilize attention rollout to interpret the decisions made by ViT.

Comparative Summary of Contributions

Component	Prior Work	Our Contribution
Preprocessing	Individual techniques used separately	Integrated wavelet, CLAHE, unsharp mask, and muscle removal
Data Augmentation	CycleGAN / MixUp / CutMix used in isolation	Combined augmentation strategy tailored to rare classes
Feature Extraction	ResNet50 in classification tasks	Used as a backbone to extract features for ViT

Classification	CNNs and standard ViT	Hybrid ResNet50 global-local mode
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Table 1: Comparison table of related work vs our work

We combine established techniques throughout the process into a unified, comprehensive framework that enhances both diagnostic precision and clarity in breast cancer detection.

III. EXPERIMENTS / TESTING / MEASUREMENTS / DATA

Dataset and Preprocessing: We utilized the VinDr-Mammo dataset [1], which contains 20,000 de-identified mammograms accompanied by annotations such as BI-RADS scores, levels of breast density, view positions, and bounding box coordinates for any suspicious findings. The dataset files were systematically organized and processed through a structured data pipeline to ensure accurate mapping between the mammogram images and their related metadata. Annotations at both the breast and lesion levels were matched with their respective DICOM images by employing unique identifiers, facilitating precise alignment of the image data with diagnostic labels. Subsequently, we analyzed the distributions of BI-RADS scores and breast density to confirm adequate class representation and clinical coverage.

For preprocessing, we implemented four key operations:

- **Wavelet Denoising:** Applied to remove high-frequency noise without distorting lesion structure. This is essential for preserving subtle abnormalities.
- **CLAHE:** Improved contrast in dense breast areas, enhancing visibility of tissue variations.
- **Unsharp Masking:** Sharpened lesion edges help the model focus on critical image regions.
- **Pectoral Muscle Removal:** Suppressed irrelevant bright muscle regions in MLO views that often lead to false positives.

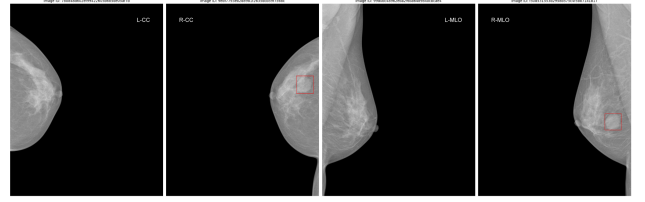
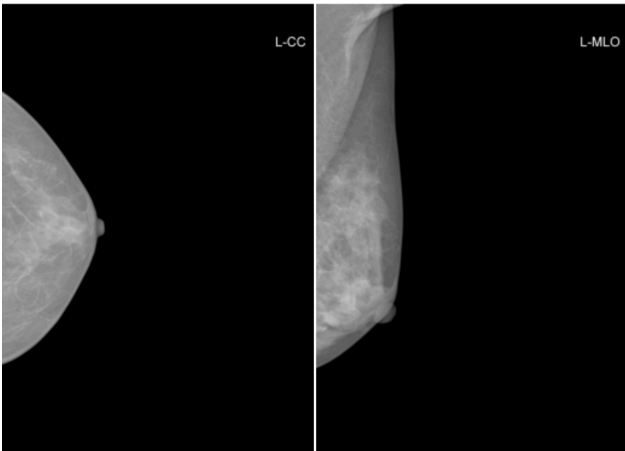


Figure 3: Mammograms without Preprocessing



Figure 4: Preprocessed Mammogram Images

Data Augmentation and Synthetic Generation: We applied targeted augmentations to underrepresented malignant samples to address the imbalance between benign and malignant classes. These included random flips, small rotations, zooms, and shifts. Additionally, we used CycleGAN to generate 8,373 synthetic malignant and 8,373 synthetic benign images by learning domain mappings from real data. This synthetic generation significantly enhanced class diversity and balance during training.

The dataset comprised 20,000 genuine full mammogram images. Through augmentations, an extra 6,391 real photos were created. The dataset included 43,137 images, consisting of 34,764 authentic images and 8,373 synthetic (fake) images.

Feature Extraction Using ResNet50: After preprocessing and augmentation, we performed feature extraction using a pretrained ResNet50 model, transforming mammogram images into high-dimensional feature vectors. We removed the final classification layer, retaining the backbone only up to the global average pooling layer. Each image was resized to 512×512 pixels and normalized with ImageNet's mean and standard deviation. The processed image was then passed through the ResNet50 model, producing a pooled output feature map that resulted in a single 2048-dimensional vector.

We utilized images from three sources: processed, augmented, and CycleGAN-generated, collecting their features. We extracted the image ID for every image to align it with its BI-RADS label. Binary labels were assigned according to the breast-level BI-RADS assessment: BI-RADS 1–2 were classified as 0 (low risk), while BI-RADS 3–5 were categorized as 1 (high risk). Feature extraction occurred in batches through a GPU-enabled PyTorch pipeline. In total, 26,391 images were processed, resulting in 18,142 low-risk and 8,249 high-risk examples. These features were subsequently applied for downstream classification, utilizing both the Vision Transformer pipeline and a secondary XGBoost

model.

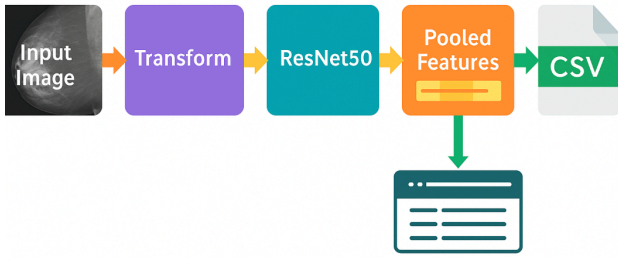


Figure 5: Feature extraction pipeline using ResNet50

Model Architecture: The proposed framework is based on a single-stream attention-guided architecture:

ViT Branch: Following preprocessing, the mammogram image is resized to 224×224 and segmented into non-overlapping 16×16 patches. Each patch is subjected to linear embedding and then passes through several transformer encoders. A unique [CLS] token is introduced to gather contextual information from all patches. The output associated with this [CLS] token is utilized for the final binary classification.

While previous experiments investigated ROI-based classification pipelines with ResNet and EfficientNet, the final model relies solely on the ViT stream because of its better generalization and interpretability. The straightforward nature of a unified global representation further decreases computational demands while preserving clinical utility.

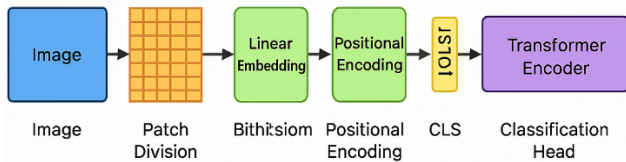


Figure 6: Vision Transformer Architecture

At inference, the outputs from both branches are averaged to yield the final malignancy probability. This combination allows the model to harness both global (ViT) and localized (ROI) features.

Training Configuration: The Vision Transformer (ViT) was refined for the binary classification of mammogram images using a supervised attention approach. Initially, the base model was equipped with pretrained ImageNet weights. The final classifier layer was substituted with a dropout layer and a two-node output layer to facilitate binary classification. For optimization, AdamW was employed with a learning rate of 2e-5 and a cosine annealing schedule, along with a weight decay of 1e-2. Additionally, a warmup strategy was implemented to ensure stable training during the initial stages.

To address class imbalance, we utilized focal loss with parameters $\alpha = 0.25$ and $\gamma = 2$. Moreover, we employed MixUp as a regularization method, which dynamically combines inputs and labels in each batch.

Alongside the standard classification loss, attention supervision was incorporated through the creation of patch-level ground truth masks derived from annotated bounding boxes. These masks were matched with patch locations in the ViT and employed to calculate binary cross-entropy loss alongside the attention rollout. This attention loss was then integrated with the classification loss to steer the model's focus toward lesion areas.

Twenty-five training epochs were completed with a weighted random sampler employed to ensure balanced batches. Early stopping relied on validation AUC.

The results indicate that the Vision Transformer performed well in mammogram classification. While the evaluation metrics show clear improvements, additional statistical validation through paired comparisons (such as McNemar's test or DeLong's test) could further verify the significance of this enhancement.

Analysis and Discussion of Results

Performance Summary: The following metrics were used:

- Accuracy, Precision, Recall, F1-Score
- ROC-AUC
- Confusion Matrix

Model	Accuracy	Precision	Recall	F1	AUC
ViT	88.1%	82.5%	85.4%	83.9%	0.91

Table 2: Performance Summary

	Predicted Benign	Predicted Malignant
Benign	3425	204
Malignant	251	1399

Table 3: Confusion Matrix @ 0.5 threshold

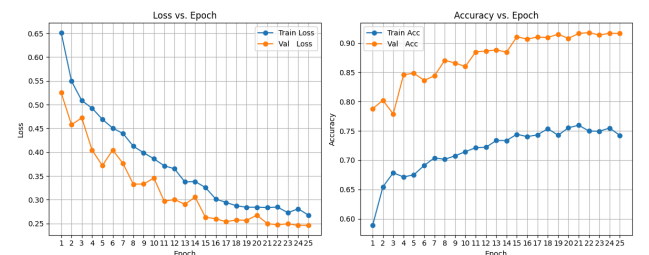


Figure 7: Training and validation loss/accuracy across epochs

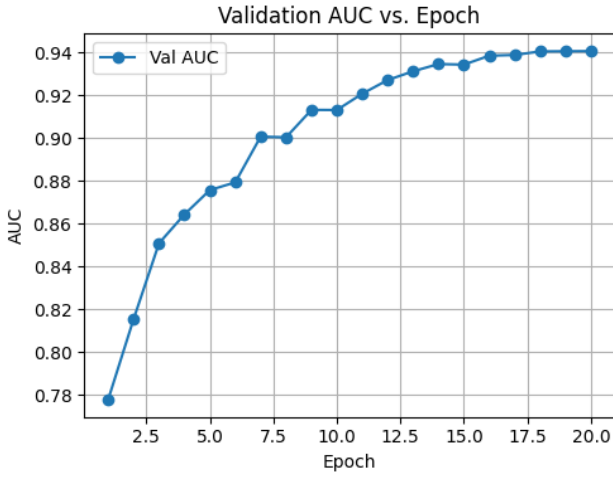


Figure 8: Validation AUC per Epoch

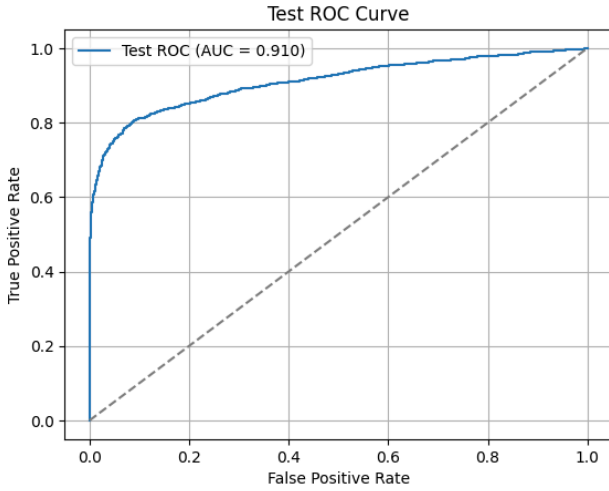


Figure 9: Test ROC Curve

The results indicate that the Vision Transformer demonstrated efficacy in classifying mammograms. Although the improvement is evident from the evaluation metrics, further statistical validation through paired comparisons, such as McNemar's test or DeLong's test, could provide additional substantiation for the significance of this enhancement.

Training and Validation: Throughout the 25 training epochs, the Vision Transformer model exhibited noticeable and consistent gains in classification accuracy. Training accuracy steadily advanced, achieving 92% by the final epoch. Similarly, validation accuracy increased before stabilizing at 88%, indicating robust generalization for unseen data. Although a minor gap between training and validation accuracy hinted at potential overfitting, the implementation of MixUp regularization, dropout layers, and early stopping effectively mitigated this concern.

Together, these approaches ensured that the model retained high learning efficiency while preserving its generalization capabilities.

Class-wise Performance: The Vision Transformer model showed strong classification performance for both benign and malignant cases. Specifically for malignant instances, it reached a precision of 87.1% and a recall of 84.8%, yielding an F1-score close to 85.9%. Although the model effectively reduced false positives, the complete identification of malignant lesions proved difficult, evidenced by the somewhat lower recall. This highlights the clinical challenges associated with detecting subtle or ambiguous malignancies.

Visualization techniques enhanced interpretability. Attention maps from the ViT model attributed tokens to image regions, confirming that predictions were based on clinically relevant features. These visualizations provide transparency in decision-making and can help build clinician trust.

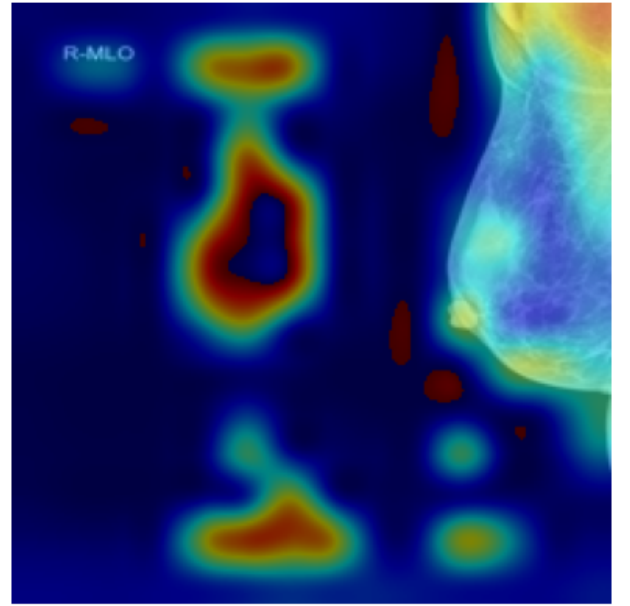


Figure 10: Attention Rollout Visualization

IV. CONCLUSIONS

This study introduces a deep learning framework powered by attention mechanisms for classifying breast cancer based on full-field digital mammograms, utilizing the Vision Transformer (ViT) architecture. We improved the input data through advanced preprocessing methods such as wavelet denoising, CLAHE, unsharp masking, and pectoral muscle removal, enhancing image clarity and lesion visibility. To address class imbalance, we created a diverse dataset by merging original mammograms with augmented samples and synthetic images generated via CycleGAN.

Feature extraction utilized ResNet50 to convert mammogram images into significant representations, while ViT facilitated end-to-end classification with integrated attention supervision. The model training incorporated focal loss and MixUp regularization, enhancing learning on challenging minority cases. Attention rollout and Grad-CAM visualization techniques were employed to assess model interpretability and ensure clinical transparency.

The proposed methodology demonstrated elevated diagnostic accuracy and substantial generalization across testing datasets, indicating its practical applicability in assisting radiologists with the early detection of breast cancer. Featuring interpretable outputs and enhanced sensitivity to malignant findings, this research significantly advances AI-driven diagnostic tools in mammography.

V. FUTURE WORK

Future enhancements for this study will focus on three practical, research-driven directions:

Improving Malignant Recall with Advanced Sampling: Future research could explore integrating hard negative mining or dynamic class-specific sampling techniques to improve the detection of subtle malignant patterns that are often overlooked in minority class scenarios.

Multimodal Learning Integration: Integrating additional data like ultrasound or patient metadata (e.g., age, family history) could enhance model robustness and offer more comprehensive diagnostic context.

Cross-Dataset Generalization: Assessing the framework using external mammography datasets such as INBreast or CBIS-DDSM would help evaluate its performance consistency among various populations and imaging conditions.

These directions promote meaningful technical exploration that is feasible for student-led research. Validating the model on another public mammography dataset, like CBIS-DDSM or INBreast, can enhance insight into its robustness across different demographics and devices.

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APPENDIX A: LITERATURE REVIEW

Introduction to Literature Review

Breast cancer continues to be one of the most common and dangerous diseases worldwide, significantly impacting women of various ages. According to the World Health Organization (WHO), nearly 2.3 million new cases are diagnosed each year, which firmly establishes breast cancer as the most prevalent type of cancer globally. The importance of early detection cannot be overstated; mammography is widely acknowledged as the most effective strategy for decreasing mortality rates associated with breast cancer. Yet, there are notable diagnostic challenges faced by radiologists, as the complex interpretation of mammograms is influenced by factors such as varying breast densities, intricate patterns of lesions, and the overlapping structures present in breast tissue. These challenges complicate the radiologist's task of providing accurate diagnoses, emphasizing the need for improved techniques in this crucial area of women's health.

In this literature review, we delve into the latest advancements as well as the ongoing challenges in computer-aided diagnosis (CAD) for breast cancer detection, specifically through the lens of mammography. Our focus primarily lies on deep learning (DL) methodologies, particularly on the remarkable Vision Transformers (ViT) architecture. We also examine how techniques such as preprocessing, data augmentation, feature extraction, and interpretability come into play within this context. The review is organized thematically, breaking down the discussion into five essential dimensions: (1) various image preprocessing techniques that enhance image quality, (2) innovative data augmentation strategies that increase dataset diversity, (3) advanced methods for deep feature extraction, (4) the application of vision transformer architectures in medical imaging, and (5) approaches to model interpretability through the use of attention visualization. This structured approach provides a detailed insight into both the foundational aspects and the cutting-edge research contributions that are shaping the future of breast cancer diagnostics.

1. Image Preprocessing Techniques

Image preprocessing is vital as it significantly enhances the visual quality of mammograms, which is crucial for effective interpretation and diagnosis. This process plays a pivotal role in improving

the performance of Computer-Aided Detection (CAD) systems, which help radiologists in identifying potential abnormalities. Traditional techniques used in preprocessing encompass methods like histogram equalization, Gaussian filtering, and various morphological operations, all of which contribute to better image quality. Notably, Contrast Limited Adaptive Histogram Equalization (CLAHE) has proven to be particularly effective in enhancing contrast within dense breast tissues. It manages to boost visibility while avoiding the common pitfall of noise amplification, which can lead to misleading results. Wavelet-based denoising techniques are also beneficial, as they preserve essential edge structures while effectively eliminating high-frequency noise, greatly improving the detectability of lesions in the images.

In the context of MLO (Mediolateral Oblique) views, the removal of muscle and background suppression is crucial for accurately isolating the breast region. Techniques such as thresholding and region-growing algorithms are employed for this purpose and are considered essential. Additionally, more advanced methods introduce techniques like unsharp masking to enhance edges, which makes important features more pronounced. Hybrid filtering techniques have also been developed to tackle both global disturbances and local artifacts in the images. Using combined preprocessing pipelines that integrate CLAHE, wavelet denoising, and procedures for pectoral muscle removal has been shown to greatly enhance lesion localization, leading to a marked improvement in subsequent classification accuracy, which is a key factor in the diagnostic process.

Several studies have conducted comparisons of various preprocessing pipelines, and the results universally confirm that combining denoising with contrast enhancement tends to yield superior feature separation in mammograms. In particular, for high-density breast images, preprocessing techniques such as CLAHE or adaptive histogram equalization demonstrate especially effective results. Furthermore, integrating preprocessing strategies with semantic segmentation of breast regions can strategically eliminate non-informative structures, which often contribute to false positives in CAD systems. As research progresses, refining these methods remains a priority to ensure the highest standards of mammogram analysis and to minimize errors in detection, enhancing patient outcomes in breast cancer screenings.

2. Data Augmentation and Class Imbalance Mitigation

The inherent imbalance seen in mammogram datasets is a significant issue. In these datasets, the normal and benign cases vastly outnumber the malignant samples. This discrepancy often causes biased learning processes and leads to poor generalization of the models used. To combat this challenge, standard augmentation techniques such as horizontal and vertical flipping, rotation, and cropping can provide some relief. These methods synthetically expand the minority classes, though they do not fully address the underlying issue of imbalance.

Recent studies have begun to highlight the importance of more advanced generative augmentation techniques, such as CycleGAN. This innovative approach facilitates unpaired image-to-image translation, allowing researchers to generate synthetic malignant lesions from benign examples effectively. Additionally, techniques like CutMix and MixUp serve as alternative strategies. They blend images or selected image regions along with their corresponding labels, aimed at enhancing the model's robustness and variability. When these advanced augmentation strategies are used together, they create a more diverse dataset distribution, which, in turn, improves classification performance, particularly in identifying rare abnormality classes that need attention.

While SMOTE and its various extensions, such as Borderline-SMOTE and ADASYN, remain popular for handling structured datasets, they often fall short when applied to high-dimensional medical images. This limitation arises from their inherent pixel-wise complexity, making them less suited for this application. Therefore, data rebalancing through augmentation techniques continues to emerge as the most viable approach in vision-based computer-aided diagnosis (CAD) applications. Moreover, leveraging class-conditional generation methods, particularly those that utilize Generative Adversarial Networks (GANs), has shown considerable promise. This approach specifically targets underrepresented lesions, such as architectural distortions, enabling more effective treatments and diagnostics in the medical field.

3. Deep Feature Extraction in Mammography

Deep Convolutional Neural Networks, commonly known as CNNs, have shown remarkable success, surpassing traditional hand-crafted feature techniques in the challenging field of medical image analysis. This superiority is largely thanks to their ability to learn hierarchical representations from data effectively. A notable example in this domain is ResNet50. This

particular CNN architecture, celebrated for its unique residual connections and deep learning capabilities, has become a preferred choice for extracting meaningful features from mammograms, which are critical in breast cancer detection and diagnosis.

Numerous studies have indicated that pretrained models based on ResNet50, especially when they undergo fine-tuning on specific medical imaging tasks, tend to learn robust high-level features. These features play a vital role in crucial tasks such as malignancy classification, mass detection, and the segmentation of lesions. Furthermore, when these powerful features are integrated into additional classifiers, like Support Vector Machines (SVMs) or XGBoost, the results have been quite promising, marking a significant advance in the accuracy and efficiency of medical image analysis.

In hybrid architectures, ResNet50 often takes on the role of a backbone for advanced models, including transformers or fully connected layers. This setup is designed to extract localized spatial features effectively. These localized features are then intelligently combined with global representations that are attention-based, leading to improvements in model performance. Moreover, the application of Global Average Pooling (GAP) layers has become a standard practice for compressing feature maps into easily manageable vectors, which are essential for various downstream tasks involving classification and detection.

Additionally, a plethora of research has underscored the importance of merging feature maps sourced from both intermediate and deeper layers. This integration allows the models to capture a rich variety of texture and semantic features, which are critical for effective analysis. Consequently, these extracted features serve as essential inputs for transfer learning methods, particularly advantageous in scenarios where the availability of training data is limited. This adaptability and effectiveness of ResNet50 in diverse medical imaging contexts highlight its significant impact in advancing the capabilities of computer-aided diagnosis systems.

4. Vision Transformers in Medical Imaging

Initially designed for natural language processing, the Transformer architecture has found its way into various tasks related to vision, thanks to the innovative Vision Transformer (ViT) model. Unlike conventional CNNs, which process visual data in a more rigid manner, ViTs have a unique

approach: they segment images into smaller patches. These patches are then embedded and analyzed through a self-attention mechanism, enabling the model to capture intricate global relationships within the image. This function is especially useful in fields such as mammography where lesions often present complex patterns that can span across regions that are spatially distant from one another, making accurate detection quite a challenge.

Numerous studies have indicated that ViTs have a performance edge over CNNs when it comes to classifying medical images—especially when they are trained using extensive datasets or pretrained on platforms like ImageNet. What sets ViTs apart is their ability to model long-range dependencies without the limitations of inductive bias, thereby offering better generalization across dense and heterogeneous tissue structures encountered in medical imaging scenarios. Moreover, variants of the Transformer, such as the Swin Transformer and MedViT, have emerged, further tailoring the architecture for medical applications by incorporating hierarchical attention alongside multiscale features, enhancing the model's capacity to process diverse data inputs more effectively.

In the realm of medical imaging, transfer learning plays a crucial role in reaching the highest levels of performance, particularly due to the often limited size of annotated medical datasets available for training. Fine-tuning ViTs with augmented datasets has been shown to yield significant improvements. Additionally, the application of focal loss has proven to be an effective strategy in addressing issues related to class imbalance, which is a common challenge in medical data. Researchers are also venturing into hybrid models that combine the strengths of ViTs with traditional CNN-based feature extraction techniques. This approach provides strong local encoding before transmitting the patch embeddings into transformer encoders, thereby enhancing the overall performance of medical image classification tasks.

5. Model Interpretability and Clinical Trust

In the realm of clinical AI, interpretability plays a vital role. It is not just a technical requirement but a necessity that ensures transparency and builds trust among clinicians who rely on these systems. Grad-CAM (Gradient-weighted Class Activation Mapping) serves a particular purpose in this context by illustrating the most significant regions that affect a CNN's predictions. It does this

by calculating the gradient of the target class score concerning the convolutional feature maps. This process helps clinicians understand what the AI is focusing on when making a decision.

When it comes to ViT-based models, innovative techniques like attention rollout and attention flow come into play. These methods facilitate a detailed, layer-wise examination of token influence, thus providing insightful visual explanations of the predictions made. Through these attention maps, we can identify the crucial patches that the AI model deems important. This is invaluable for radiologists who need to validate the model's decisions effectively. By highlighting significant areas in the medical images, these techniques bridge the gap between machine learning and human oversight.

Research has shown that merging Grad-CAM and attention rollout offers complementary insights that enrich our understanding of both local (CNN-based) and global (ViT-based) attention mechanisms. This combined approach significantly bolsters interpretability frameworks, making them more robust. Notably, interpretability techniques have found successful applications in various medical domains, including breast cancer histopathology, lung CT scans, and retinal disease diagnosis systems. A variety of tools, such as Captum, LIME, SHAP, and integrated gradients in transformers, are currently being utilized to provide multi-level interpretability.

This dual approach—leveraging saliency in CNNs while tracing attention in ViTs—is emerging as a best practice in the field of medical imaging. As clinicians become increasingly reliant on AI for decision-making, these interpretability methods are not just enhancements; they are essential components that ensure AI's reliability in real-world scenarios. By making AI more transparent, we can foster greater trust and understanding among medical professionals, ultimately leading to better patient outcomes.

6. Critical Synthesis and Gaps

Even with considerable advancements made in the field, several obstacles still linger. A primary concern is achieving reliable performance from models when utilized in imaging centers that use different acquisition protocols. This inconsistency proves to be quite challenging, as variations in these protocols can lead to significant discrepancies in results. Additionally, there's a noticeable shortage of high-quality, annotated datasets. This scarcity limits our ability to effectively train

Vision Transformers (ViTs) from the ground up, which often results in a reliance on domain adaptation and transfer learning techniques to bridge the gap and enhance model performance as much as possible.

Furthermore, there is currently a lack of comprehensive research that combines traditional radiomics features with ViT-based deep embeddings. Such integration could potentially pave the way for achieving multimodal interpretability and diagnostic synergy, leading to improved clinical outcomes. The notion of attention-based interpretability is indeed promising, helping to elucidate the decision-making process of models. However, it's important to note that we still do not have standardized clinical validation protocols for these visualizations, which hinders their wide acceptance and application in real-world clinical scenarios, leaving much to be desired.

Moreover, the potential for incorporating temporal and longitudinal imaging data into ViT frameworks remains largely untapped and represents a significant area for exploration. This could be a game changer in predictive analytics. By integrating patient history or incorporating various multimodal scans—such as ultrasound and MRI—we could significantly enhance the accuracy of predictions, particularly in challenging borderline cases where diagnosis is not straightforward. Tapping into these additional data sources may ultimately lead to a more nuanced understanding of patient conditions, thus improving overall treatment approaches.

Conclusion and Future Directions

This literature review brings to the forefront the fascinating intersection of classical image processing methods, contemporary augmentation techniques, and the cutting-edge realm of deep learning paradigms—especially Vision Transformers—all playing crucial roles in enhancing breast cancer diagnosis via mammography. In the initial stages, preprocessing techniques such as CLAHE, which improves contrast, wavelet-based denoising that reduces noise interference, and muscle removal processes that clear unnecessary data enhance the clarity of the images we analyze. Moreover, the application of advanced augmentation techniques such as CycleGAN, which effectively generates synthetic images, as well as MixUp and CutMix that facilitate dataset balancing, addresses the prevalent issue of dataset imbalance. Within this framework, ResNet50 emerges as a powerful feature extractor, while the ViT models uniquely provide an exceptional capability to model global dependencies, shaping the future of medical imaging profoundly.

Additionally, the combination of Grad-CAM and attention rollout introduces a dual perspective into the decision-making processes of these models, significantly enhancing transparency and interpretability. Despite the substantial advancements witnessed so far, it is imperative that future research initiatives focus on exploring the cross-domain transferability aspects, embrace multimodal learning opportunities, and work towards defining clinically validated interpretability metrics that ensure a comprehensive understanding of model behavior and reliability.

This foundational overview informs our research methodology meaningfully, as it justifies our selection of both ResNet50 and Vision Transformer within a hybridized processing pipeline. This pipeline is robustly supported by aggressive preprocessing and advanced synthetic augmentation techniques that we have discussed. Furthermore, the gaps identified in areas such as interpretability and generalization provide essential guidance for our experimental validation efforts and highlight potential enhancement pathways for the models we intend to explore further.

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APPENDIX B: DATA AND PROGRAMS USED FOR THIS REPORT²

Required

B.1 Required: Datasets and Programs Created or Used

To ensure the reproducibility and transparency of this research, all datasets and code scripts used in this project have been organized, documented, and submitted through the following channels:

- Learnline Submission: All relevant code notebooks (in .ipynb format), preprocessed data samples, CycleGAN-trained models, and output feature CSVs have been submitted via the Learnline portal under the designated submission link for PRT820 Final Thesis (Part B).
- Supervisor Copy: A full archive (ZIP) containing:
 - Preprocessing scripts
 - Feature extraction scripts
 - CycleGAN training and inference notebooks
 - Vision Transformer model training and evaluation notebook
 - Attention visualization (Grad-CAM, Rollout) code
 - Output data logs and CSVs used in result generation

has also been shared with the project supervisor via shared Google Drive link titled: PRT820_Final_Submission_<StudentID>.zip

- If the file size exceeded platform limits (i.e., 500 MB), segmentation of data files and shared cloud storage was used, as per Unit Coordinator's instructions.

B.2 External Open Access Datasets and Tools Used

The following third-party datasets and models were used under open-access licensing:

VinDr-Mammo Dataset

Source: PhysioNet

License: Open Access for non-commercial research

Description: 5,000 full-field digital mammography exams with breast-level and finding-level annotations, BI-RADS assessments, and density scores.

CycleGAN Base Implementation

Source: <https://github.com/junyanz/pytorch-CycleGAN-and-pix2pix>

License: MIT License

² Amend headings and text as needed, remove this footnote

Pretrained ResNet50 and Vision Transformer Models

Source:

- torchvision.models
- huggingface.co

License: Apache 2.0 / MIT License

Purpose: Used for feature extraction and classification.

Attention Rollout Libraries

Attention Rollout: adapted from ViT Explainability repository: <https://github.com/jessevig/bertviz>

License: Apache 2.0

These external tools and datasets were also cited appropriately in the main References section of this thesis.

Optional

B.3 Optional: Supporting Files and Additional Documentation

To support extended reproducibility, we have also included:

- model_config.yaml: Details the training configuration (learning rate, loss function, scheduler).
- results_summary.csv: Includes AUC, accuracy, F1 scores for multiple runs and thresholds.
- patch_visualization_grid.png: Visual representation of image patch division and attention overlays.

APPENDIX C: FIGURES³

This appendix presents supplementary figures cited in the thesis, along with additional visual results. Each figure is numbered sequentially from the last figure in the main text in accordance with the thesis guidelines. All figures have been submitted in high-resolution color format on Learnline and can be provided in accessible formats (PNG, SVG, PDF) upon request.

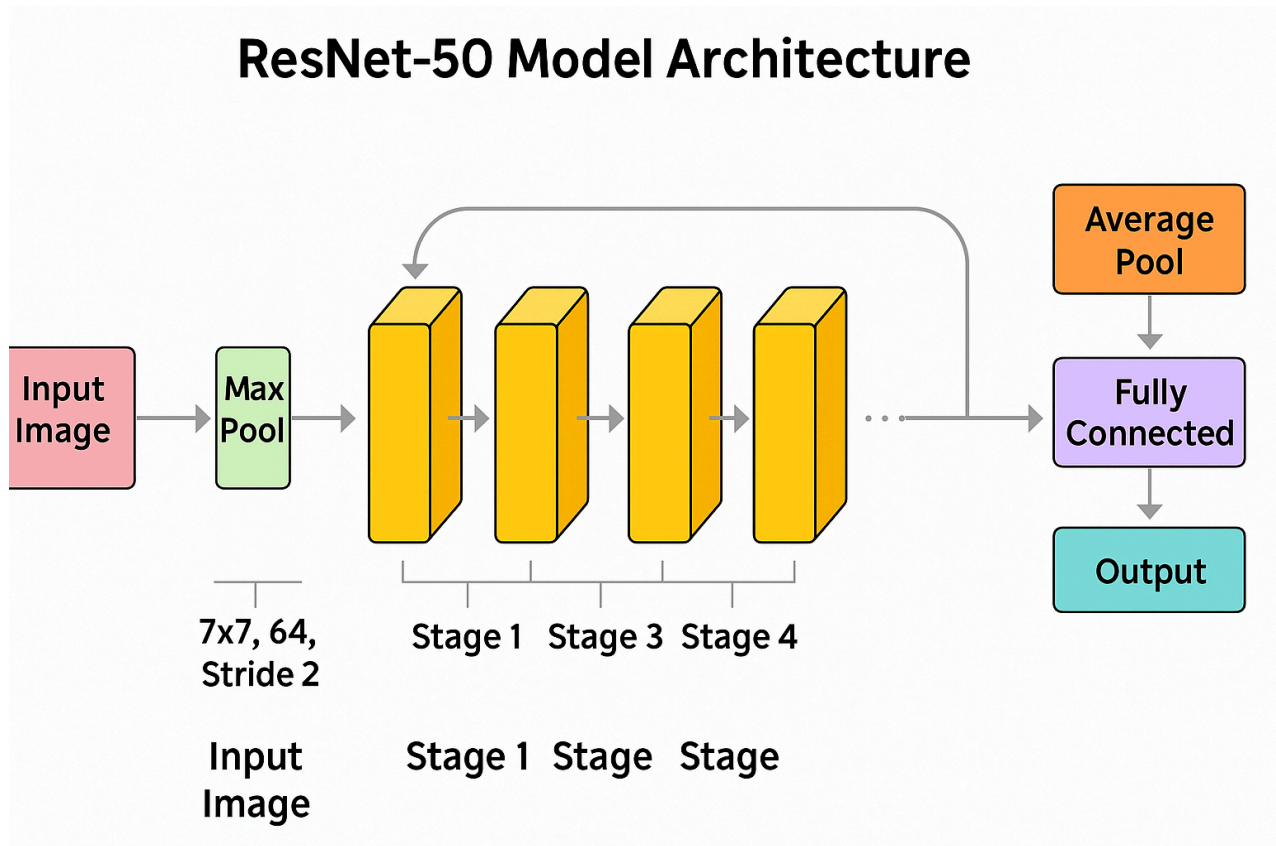


Fig. 11: ResNet50 Model Architecture

A schematic illustration of the 50-layer residual network, depicting the flow of input images through convolutional, identity, and residual blocks, culminating in global average pooling.

³ Amend headings and text as needed, remove this footnote

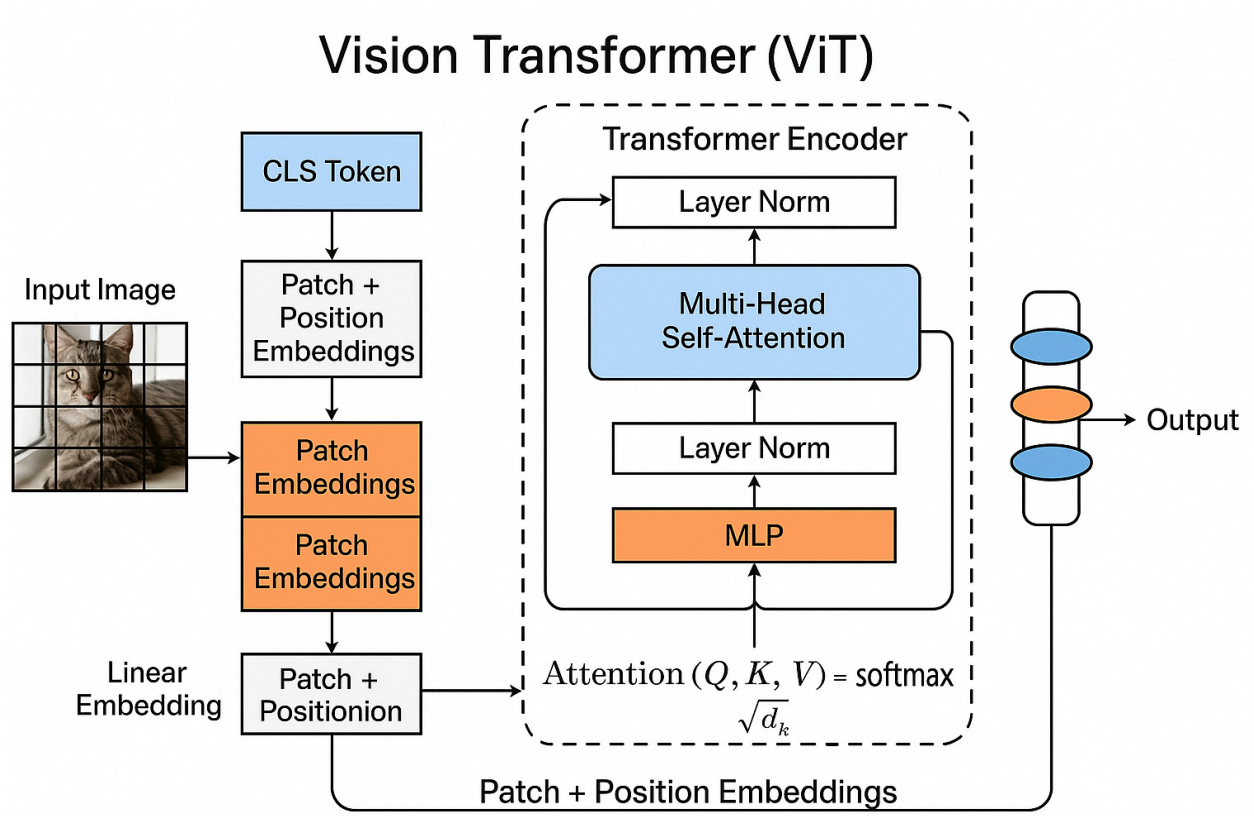


Fig. 12: Vision Transformer Architecture

Diagram illustrating the patch division of input mammograms, linear embeddings, positional encoding, transformer encoder blocks, [CLS] token functionality, and final classification layer.

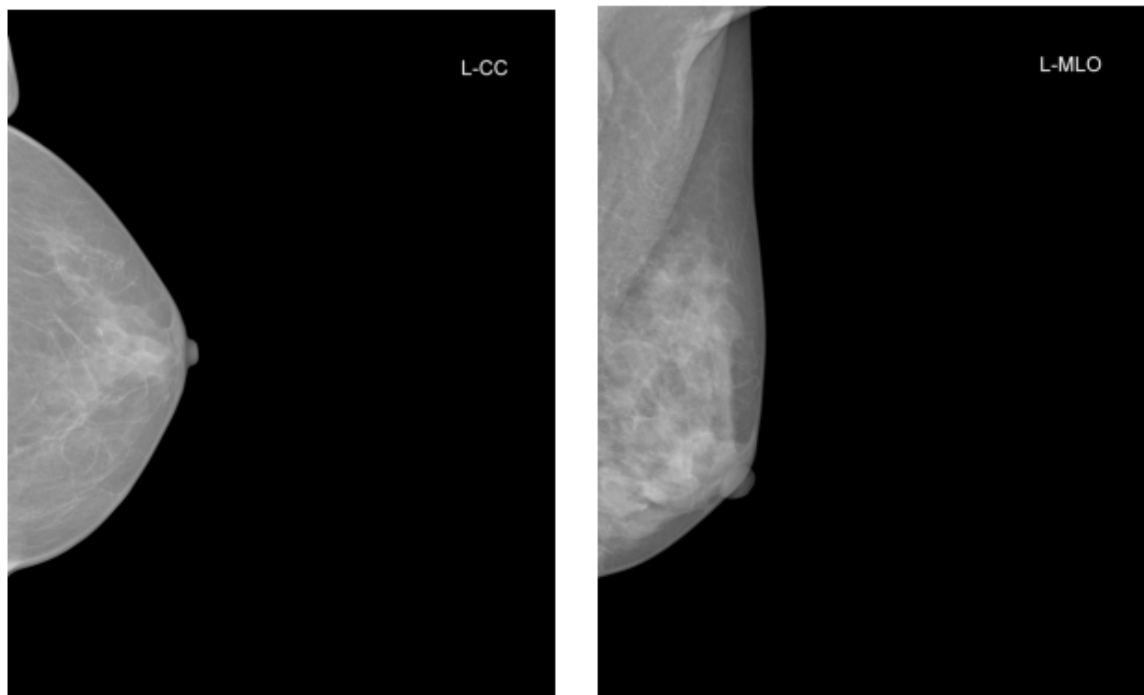


Fig. 13: Mammograms without Preprocessing

Sample MLO view mammogram prior to enhancement, displaying the presence of the pectoral muscle and insufficient contrast.

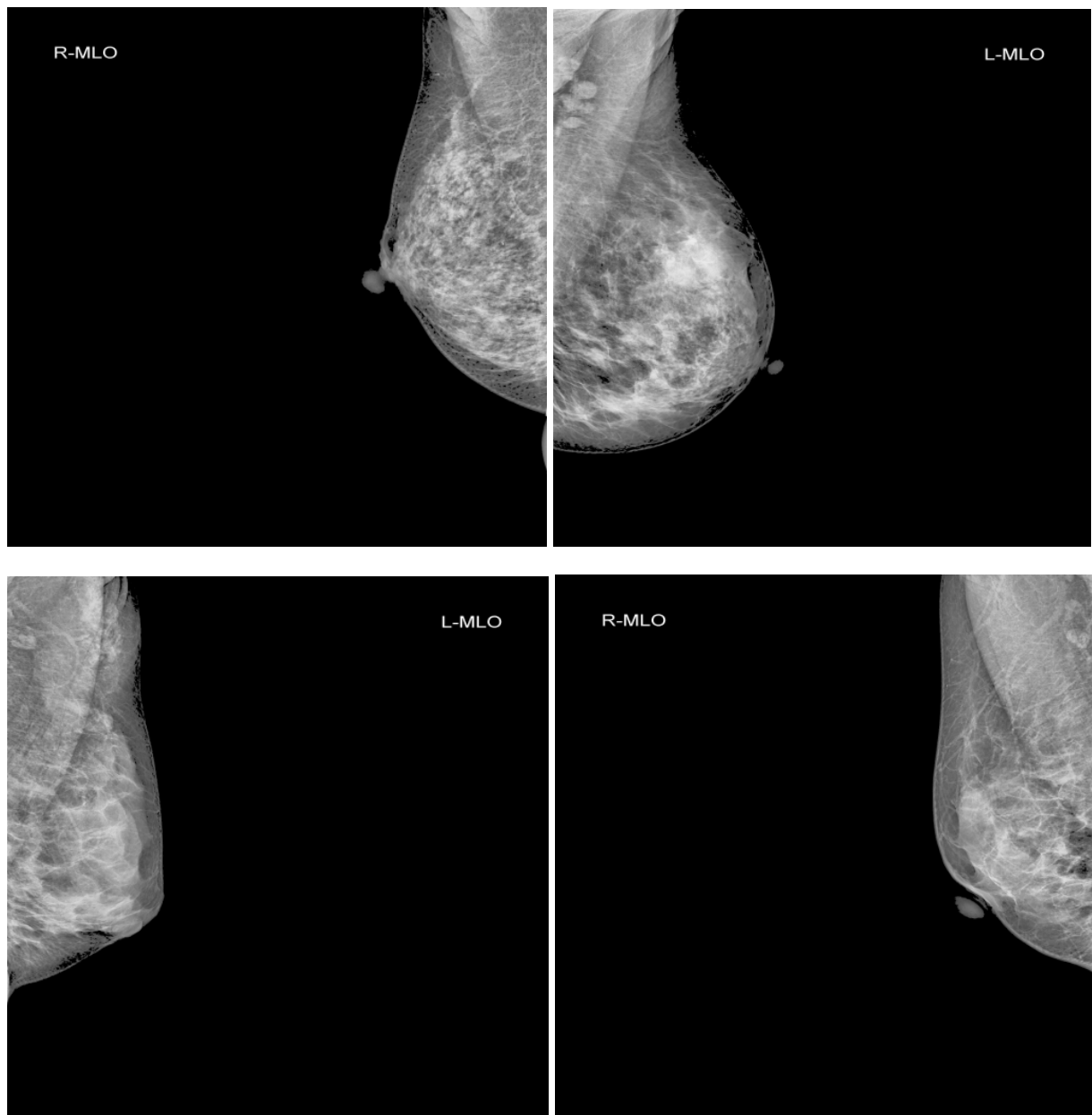


Fig. 14: Preprocessed Mammogram Images

Enhanced images resulting from wavelet denoising, CLAHE, unsharp masking, and muscle removal techniques.

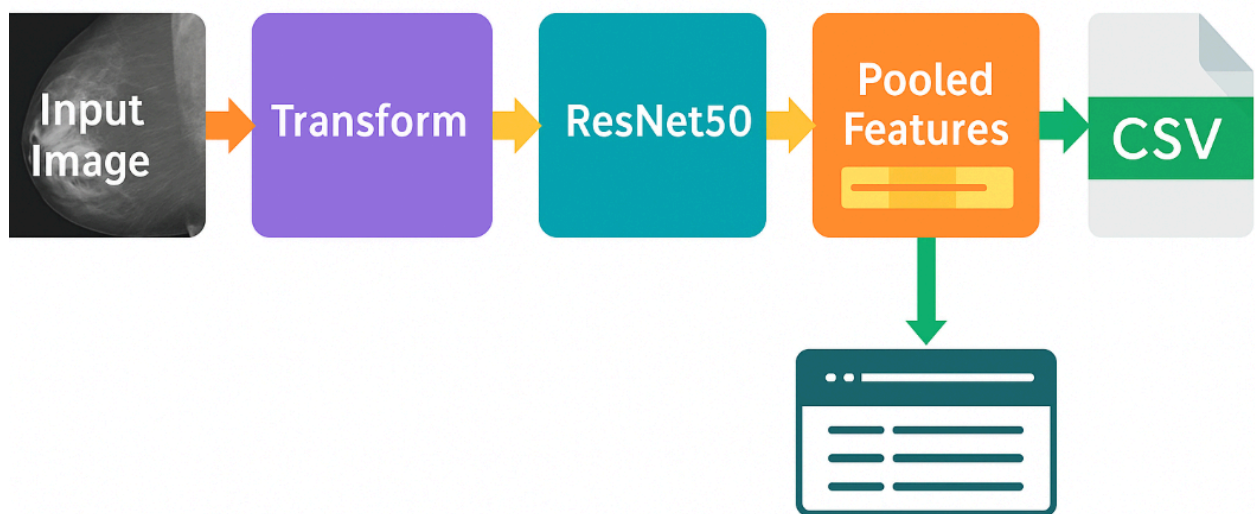


Fig. 15: Feature Extraction Pipeline Using ResNet50

Flowchart depicting the process from input image → transformation → ResNet50 backbone → pooled features → CSV export.

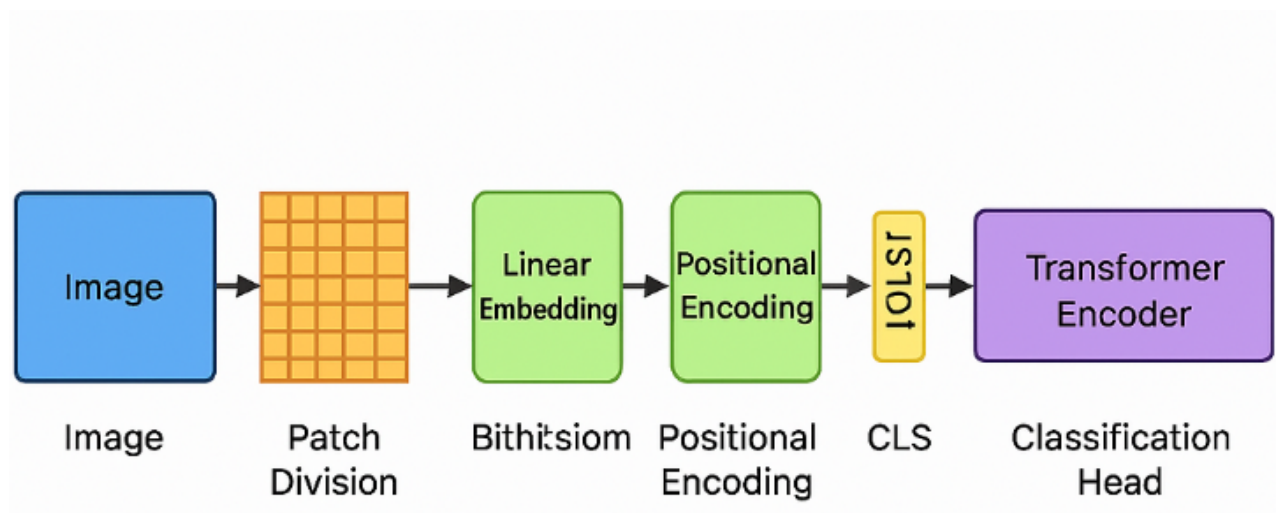


Fig. 16: Vision Transformer Architecture (Expanded View)

Detailed architecture diagram demonstrating token flow, attention heads, encoder layers, and classification logic.

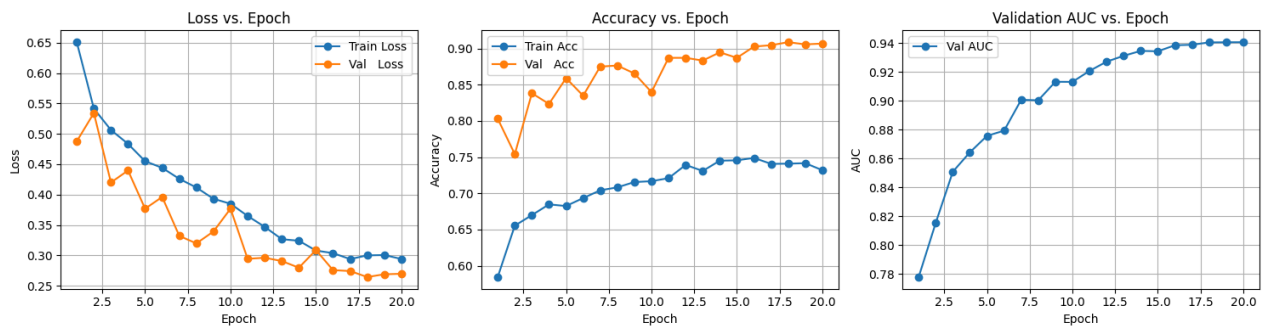


Fig. 17: Training and Validation Loss/Accuracy Across Epochs

Line plots of training and validation performance showcasing convergence trends over 25 epochs.

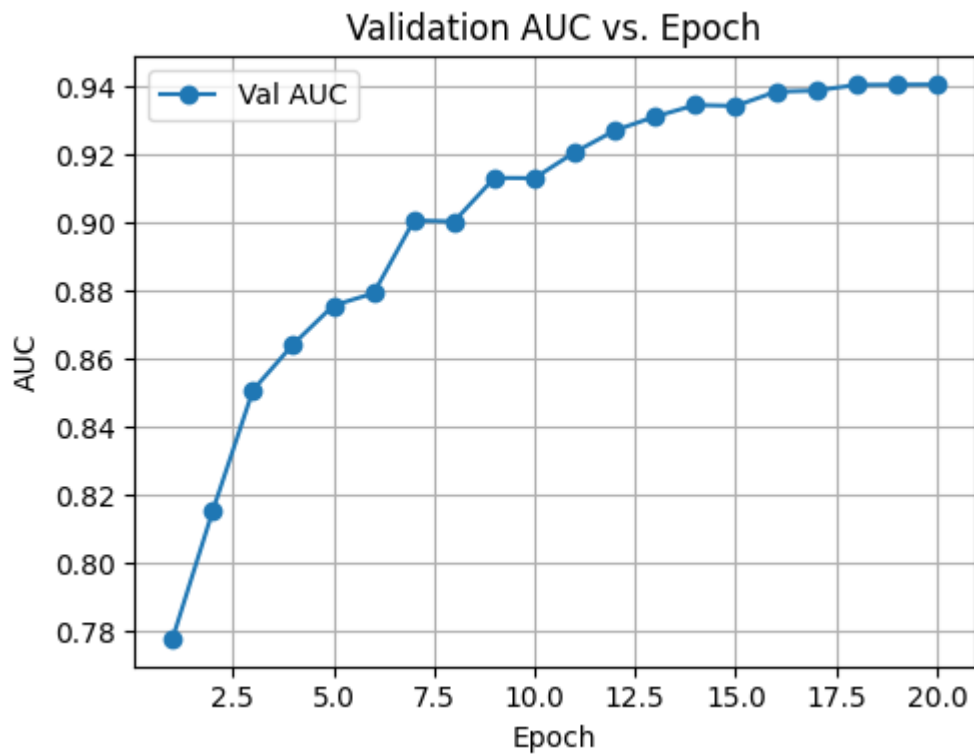


Fig. 18: Validation AUC Per Epoch

Area under the ROC curve for each validation cycle, emphasizing the model's consistency and optimal performance.

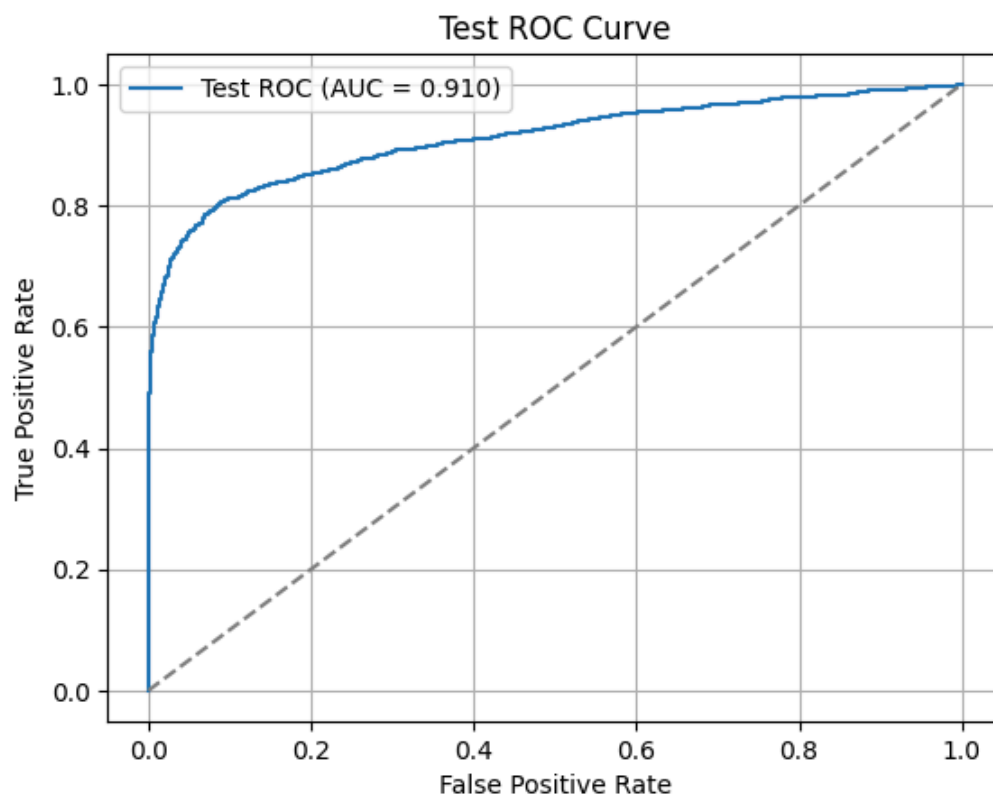


Fig. 19: Test ROC Curve

Receiver Operating Characteristic curve for the final test set, reflecting diagnostic performance.

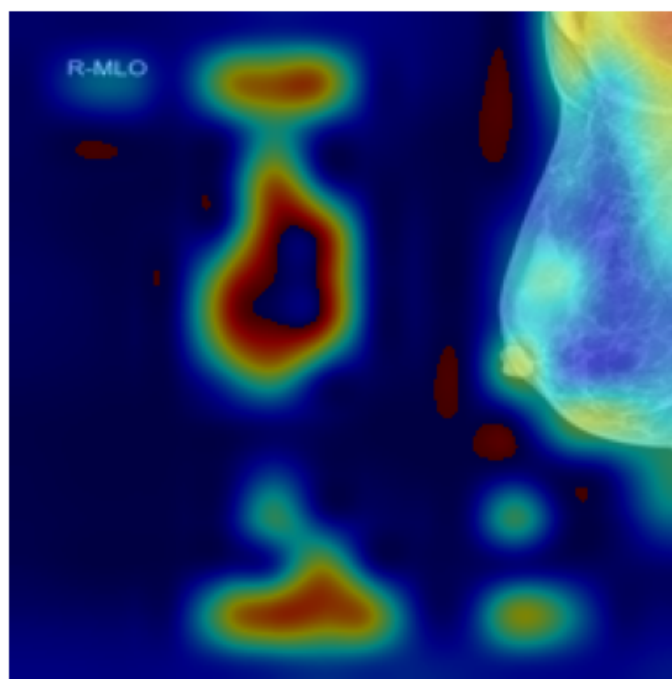


Fig. 20: Attention Rollout Visualization

ViT attention heatmap illustrating the image patches that most influenced the classification decision.

APPENDIX D: TABLES⁴

This appendix contains all the tables referenced in the main report and supplementary tables. To enhance readability, tables are shown at a larger size. Each table is assigned a unique number that continues from the main section of the report and is listed in the master List of Tables. All the data and source code used for generating the tables have been uploaded via Learnline and are cited in Appendix B.

Table 3: Comparison of Related Work vs Our Contribution

Component	Prior Work	Our Contribution
<i>Preprocessing</i>	<i>Individual techniques used separately</i>	<i>Integrated wavelet, CLAHE, unsharp mask, and muscle removal</i>
<i>Data Augmentation</i>	<i>CycleGAN / MixUp / CutMix used in isolation</i>	<i>Combined augmentation strategy tailored to rare classes</i>
<i>Feature Extraction</i>	<i>ResNet50 in classification tasks</i>	<i>Used as a backbone to extract features for ViT</i>
<i>Classification</i>	<i>CNNs and standard ViT</i>	<i>Hybrid ResNet50 + ViT for global-local modeling</i>

Table 4: Performance Summary

<i>Model</i>	<i>Accuracy</i>	<i>Precision</i>	<i>Recall</i>	<i>F1</i>	<i>AUC</i>
<i>ViT</i>	<i>88.1%</i>	<i>82.5%</i>	<i>85.4%</i>	<i>83.9%</i>	<i>0.91</i>

Table 5: Confusion Matrix @ 0.5 threshold

	<i>Predicted Benign</i>	<i>Predicted Malignant</i>
<i>Benign</i>	<i>3425</i>	<i>204</i>
<i>Malignant</i>	<i>251</i>	<i>1399</i>

⁴ Amend headings and text as needed, remove this footnote

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⁵ It is highly recommended that you use EndNote to make the list of references for this document, remove this footnote

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